Back to Nature: A Real-time System for Indonesian Medicinal Plant Identification Using Convolutional Neural Network as an Effort to Strengthen Traditional Medicine

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Abstract. Amid rapid technological advancement, there's a resurgence of interest in herbal remedies as people seek a return to nature to alleviate health issues. Indonesia's abundant medicinal plants, like turmeric and ginger, are key players in global health initiatives. Recent export data underscores the country's natural wealth and economic potential. However, public familiarity with traditional medicinal plants is limited to processed forms. To address this gap, research focuses on real-time AI-based plant detection using Convolutional Neural Networks (CNN), leveraging CNN's proven accuracy in object recognition. used k-Fold Cross Validation to validate the training results and used 5 Fold. The evaluation revealed that even the least performing model achieved a high validation accuracy of 71.144%, while the top-performing model achieved a perfect validation accuracy of 79.104%. Similarly, the weakest model demonstrated a testing accuracy of 73.50%, consistent with its validation accuracy, whereas the best model achieved an outstanding testing accuracy of 73.00%. Additionally, a web-based system has been successfully deployed using the best model, operating seamlessly. Future enhancements could focus on refining accuracy through hyperparameter tuning and expert integration into the web system.

Keywords: Medical Plants, CNN, Real Time, AI, Natural

1. Introduction

In an era of rapid technological advancement, people are increasingly yearning for familiarity with nature, leading to a resurgence of interest in herbal remedies. The slogan "Back to Nature" is not only a call, but also a symbol of the drive to reduce the negative impacts that may occur [1]. Nature is rich in a variety of medicinal plants with different active substances, providing various therapeutic effects such as relieving pain, treating hypertension, diabetes, cholesterol, and increasing endurance. Indonesia is one of the countries rich in natural resources, Indonesia leading medicinal plants such as turmeric, red ginger, temulawak, and others have long been used by the community to maintain health [2], with the hope of helping the world in the SDGs (Sustainable Development Goals) mission point 3 on Good Health and Well Being. Based on the latest data from the Central Bureau of Statistics/Badan Pusat Statistik (BPS), Indonesia

exports of medicinal plants, aromatics and spices peaked in 2022, showing an increase of 5.55% from the previous year. The total export amount of 279.3 thousand tons not only reflects Indonesia natural wealth, but also signifies its huge economic potential, with an overall value of USD607.86 billion [3].

However, in this era, it is rare for people to directly see the original form of traditional medicinal plants. Many are only familiar with medicinal plants in processed forms such as powders or pills. Therefore, with the economic potential and health benefits possessed by medicinal plants, as well as the lack of public knowledge about their original form, research was conducted to develop traditional medicinal plant detection technology based on real time images or videos. This research aims to identify the types of traditional medicinal plants and develop artificial intelligence technology that can mimic the ability of the human eye and brain in recognizing these medicinal plants.

This research uses the Convolutional Neural Networks (CNN) algorithm by looking at the results in previous studies, for example research from [4] which shows the maximum real time face recognition accuracy of 98.75% and 98.00%. Further research was conducted by [5] which showed an accuracy of 99.94% for the classification of traffic signs. Another research from [6] for hand gesture recognition classification with an accuracy of 94.61%. This shows that the CNN algorithm can produce very accurate classifications or predictions in object detection.

2. Method

2.1 Dataset

In this research, the focus is on four types of medicinal plants (Figure 1) based on their leaves that provide invaluable health benefits.



(a) (b) (c) (d)

Figure 1. four types of medicinal plants (a) ginger, (b) turmeric, (c) temulawak, (d) galangal

First is ginger (Zingiber officinale) in **Figure 1a**, which is widely proven to help boost the immune system, a feature that is especially important amid the COVID-19 pandemic [7] . Next, there's turmeric (Curcuma longa) in **Figure 1b**, which is well known for its various benefits, including its ability to aid cancer therapy [8]. Then, there's temulawak (Curcuma zanthorrhiza) in **Figure 1c**, known as a versatile medicinal plant for indigestion, jaundice, as well as maintaining overall health [9]. Then, there's galangal (Alpinia galanga) in **Figure 1d**, which is used to treat a variety of health issues, ranging from aphrodisiacs, postpartum blood flow, contraception, and rheumatism [10].

2.2 Design Research Steps

The steps in this research include (**Figure 1**):

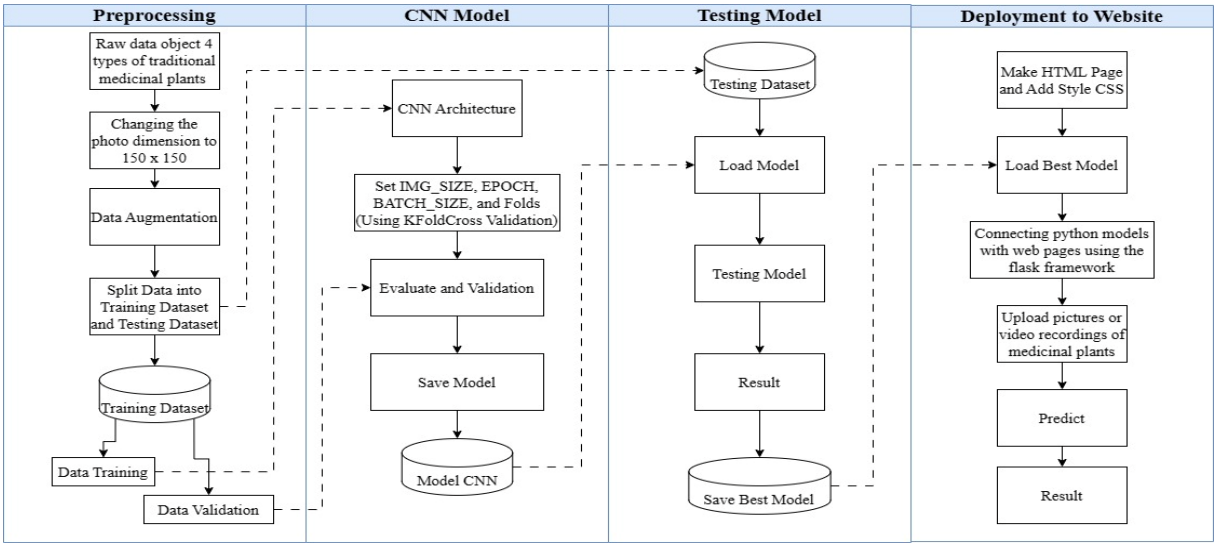


Figure 1. Research Steps

This research has several steps, starting from data collection which then enters the steps in **Figure 1** starting from the data preprocessing section, where first changing the dimensions of the photos from the collected data to 150 x 150 according to the needs of CNN, then because the collected data is limited, data augmentation is carried out to increase the data and after that the data is divided into 1004 data for training data (training dataset and validation dataset) and 200 data for testing data. Continued in the second part, namely the CNN Model where the first determines the model of the CNN architecture (**Figure 2**).

Then set the Image Size to 150, the number of epochs required for training and in this study used a number of 100 epochs and used k-Fold Cross Validation to validate the training results and used 5 Fold. Then enter the training process and after that 5 CNN models are obtained which are then stored. Then enter the model testing section to check the performance of each model with new data outside the testing dataset, the steps in this section are to prepare the testing dataset and load all models for testing, then the model with the best accuracy is saved. The next part is deployment to the website where it starts with creating a web page with HTML and CSS, then preparing the best CNN model and connecting between the web page and the CNN model using the python framework, namely Flask, then testing is done to the website by entering photos or realtime videos and the system produces the results of predicting the type of medicinal plant.

2.3 CNN Architecture

n this study using a CNN model (**Figure 2**) with an architecture starting with the first convolution layer consisting of 16 filters with a size of 3x3 pixels, which is applied with the Rectified Linear Unit (ReLU) activation function.

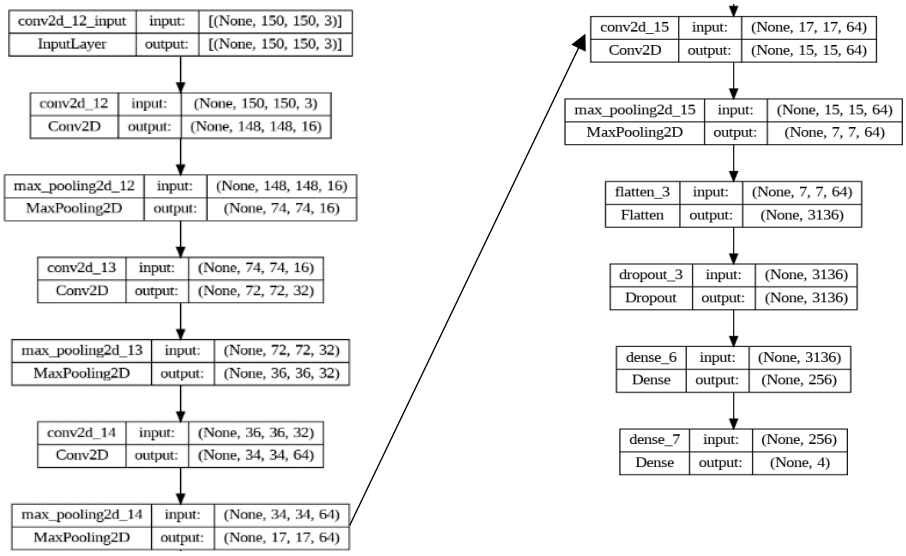


Figure 2. CNN Architecture

Next, the result of this convolution goes through a 2x2 max pooling layer to reduce image size and improve computational efficiency. This process is repeated on the second convolution layer with 32 filters and ReLU activation, followed by another 2x2 max pooling layer. The third convolution layer follows the same pattern with 64 filters and ReLU activation, which is followed by another 2x2 max pooling layer. The last step in the convolution process is the fourth convolution layer with 64 filters, ReLU activation, and 2x2 max pooling layers. Once the convolution process is complete, the output of the last layer is flattened into a one-dimensional vector to be prepared for the next layer. A dropout layer is then used with a dropout rate of 0.5 to reduce overfitting by randomly ignoring some units during training. The result vector goes through two Dense (fully connected) layers, with the first layer containing 256 neurons and ReLU activation, and the second layer containing 4 neurons (corresponding to the number of desired classes) with softmax activation to generate output class probabilities.

2.4 Confusion Matrix

Confusion Matrix serves as a tool to assess the effectiveness of the model in classification tasks [11]. **Table 1** shows the confusion matrix with four classification classes of medicinal plants Zingiber officinale, Curcuma longa, Curcuma zanthorrhiza, and Alpinia galanga.

Table 1. Confusion Matrix with Four Classification Classes Of Medicinal Plants

		Actual Values			
		Zingiber officinale	Curcuma longa	Curcuma zanthorrhiza	Alpinia galanga
Predicted Values	Zingiber officinale	T	F	F	F
	Curcuma longa	F	T	F	F
	Curcuma zanthorrhiza	F	F	T	F
	Alpinia galanga	F	F	F	T

The formulas used to calculate accuracy Equation (1), precision Equation (2), recall Equation (3), and F-1 Score Equation (4), based on [12], are:

$$\text{Accuracy} = \frac{\text{Total Correct Data}}{\text{Total Test Data}} \times 100\% \quad (1)$$

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} \quad (2)$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} \quad (3)$$

$$\text{F-1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

3. Result and Discussion

3.1 Training and Validation

The training process was conducted with the parameters epoch = 100, batch size = 32, and 5 times of cross validation were performed. **Table 2** shows the loss and accuracy of the validation results for five folds, while **Figure 3** shows the accuracy and loss graphs for fold-1. **Figure 4** for fold-2, **Figure 5** for fold-3, **Figure 6** for fold-4, and **Figure 7** for fold-5.

Table 2. Loss and accuracy values of validation results

Fold	Accuracy (%)	Loss	Validation (Photo)	
			True	False
Fold-1	76.119	0.596	153	48
Fold-2	71.144	0.572	143	58
Fold-3	75.621	0.456	152	49
Fold-4	79.104	0.422	159	42
Fold-5	73.500	0.403	147	53

Fold-4 produces the most accurate model, whereas fold-2 yields the most miniature accurate model despite its sufficient accuracy at 71.144% and the best model is generated from fold-4.

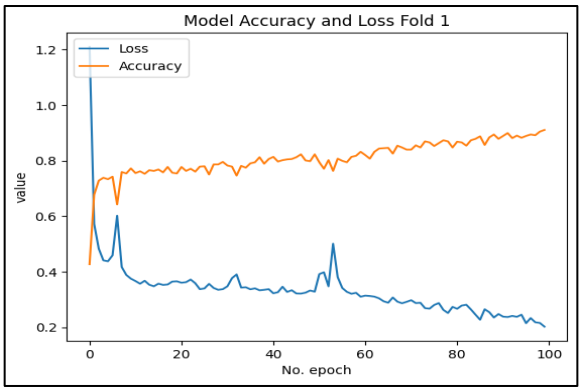


Figure 3. Accuracy and Loss for Fold-1

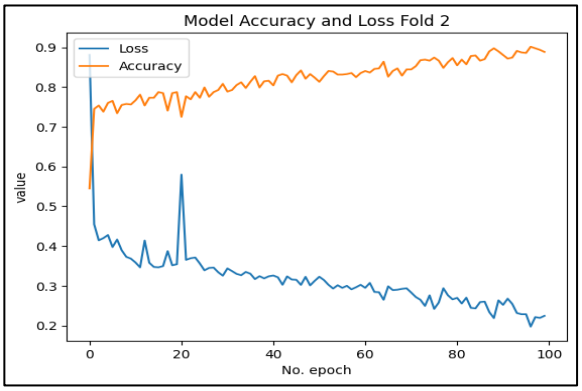


Figure 4. Accuracy and Loss for Fold-2

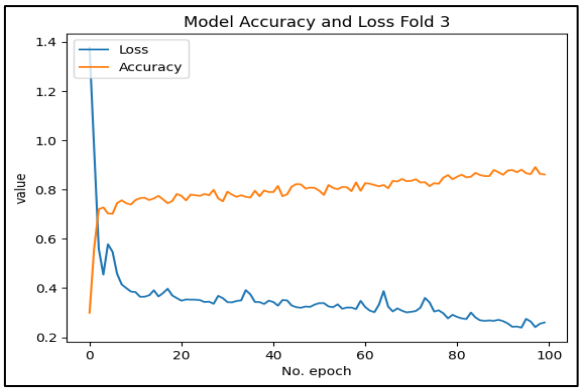


Figure 5. Accuracy and Loss for Fold-3

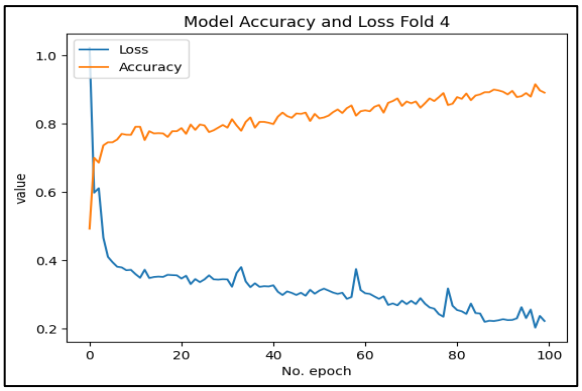


Figure 6. Accuracy and Loss for Fold-4

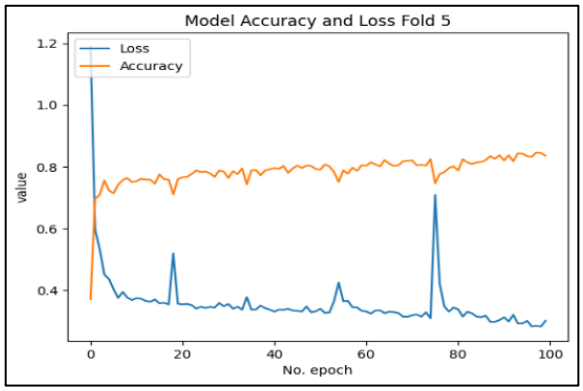
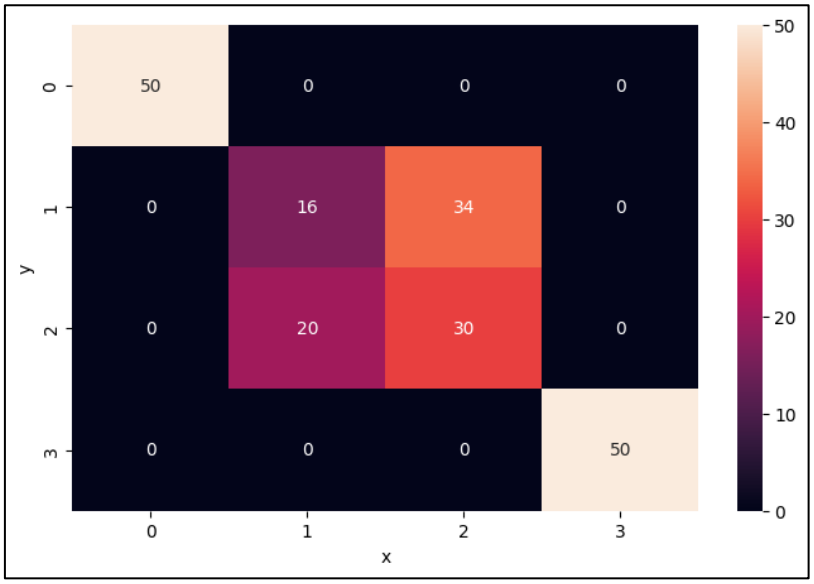


Figure 7. Accuracy and Loss for Fold-5

3.2 Testing

The testing process in this study uses 200 testing images consisting of 50 images from 4 classes of medicinal plant images. The testing process is carried out to test the performance of the CNN model that has been obtained from the training process which has the highest accuracy (model 4). The variables to be assessed in the testing process are 'False' and 'True' of the image classification generated from CNN. The 'True' and 'False' values will be used to calculate the accuracy of the CNN model. The testing results with best model are shown in **Table 3** and the convolution matrix in **Figure 8**.

Figure 8. The Confusion Matrix of the Best model

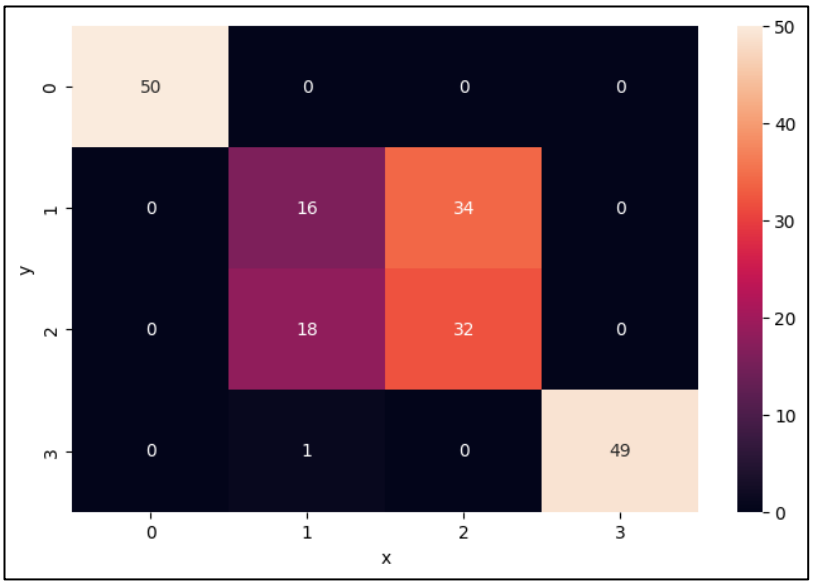


Tabel 3. Testing results of the Best CNN model

Medicinal Plant Type Class	Total Test Data	False	True	Precision	Recall	F1-Score	Accuracy (%)	Loss
Alpinia galanga	50	0	50	1.00	1.00	1.00		
Curcuma zanthorrhiza	50	34	16	0.44	0.32	0.37		
Curcuma longa	50	20	30	0.47	0.60	0.53		
Zingiber officinale	50	0	50	1.00	1.00	1.00		
Total	200	54	146				0.7300	0.4875

In the test using even the worst model in **Figure 9** and **Table 4**, the results show an improvement where the detection does not depend on how good the model is.

Figure 9. The Confusion Matrix of the worst model



Tabel 4. Testing results of the Worst CNN model

Medicinal Plant Type Class	Total Test Data	False	True	Precision	Recall	F1-Score	Accuracy (%)	Loss
Alpinia galanga	50	0	50	1.00	1.00	1.00		
Curcuma zanthorrhiza	50	19	16	0.44	0.32	0.37		
Curcuma longa	50	1	49	0.47	0.60	0.53		
Zingiber officinale	50	34	32	1.00	1.00	1.00		
Total	200	54	147				0.7350	0.3855

3.3 Deployment

Following multiple tests, the optimal value has been identified and will be incorporated into the system. The system, structured as a web-based application, is capable of identifying medicinal plants. Shown in **Figure 10** is the main menu header of the application, presenting various features, notably the model for identifying medicinal plants. Constructed on the Python-Flask framework, the application leverages this web application framework written in Python, which furnishes libraries, tools, and technologies for developing dynamic web applications. Utilizing the Python-Flask framework ensures a user-friendly interface accessible through a web browser. Ultimately, the system aims to streamline medicinal plant identification and provide valuable support to herbal medicine researchers. To facilitate the detection of images of known medicinal plants, the website offers an image upload feature. This functionality allows users to upload images of medicinal plants they want to identify. The form depicted enables users to browse and select an image of the medicinal plant they wish to detect. The model utilizes deep learning algorithms for image recognition and detection. Trained on a large dataset of diverse medicinal plant images, the model can accurately identify and classify plants based on their unique features and characteristics. The website aims to provide a straightforward and efficient means for users to identify medicinal plants and gain insights into their properties and uses through this image upload feature.

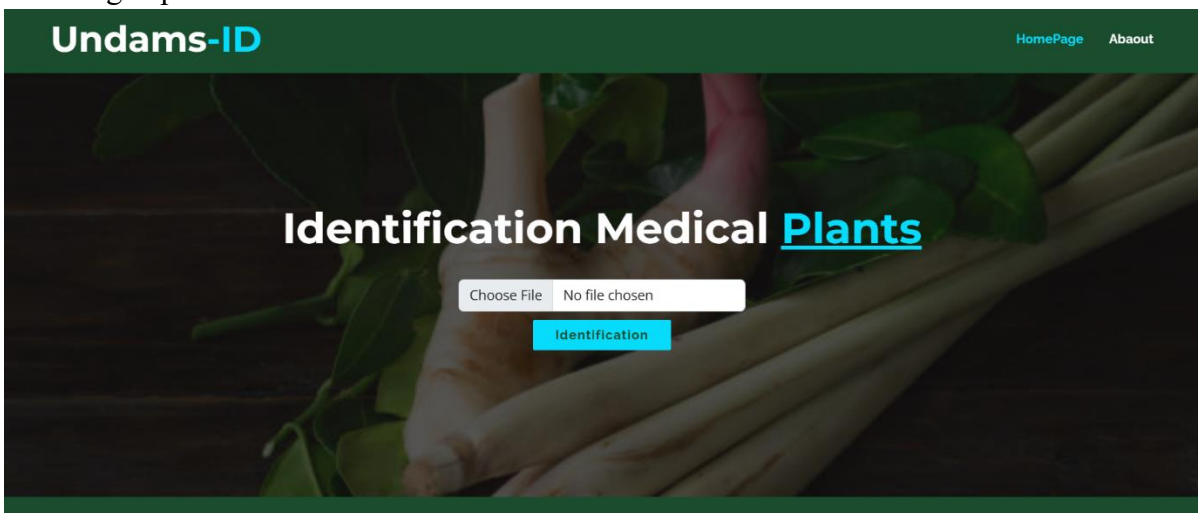


Figure 10. Main Menu of Web-Based Applications for Detect Medical Plants

The image uploaded by the user is displayed in **Figure 11**, indicating that it is ready for detection. By clicking the "Detection" button, the website will initiate processing of the image and present the detected medicinal plant along with its name and other pertinent details, including common name, scientific name, and medicinal properties, as illustrated in **Figure 11**.

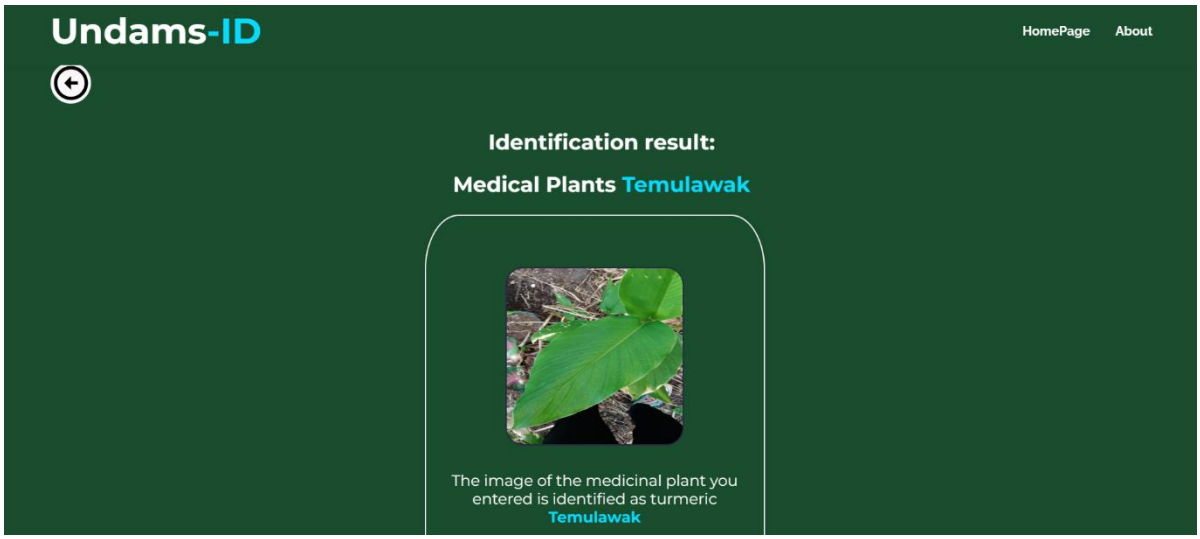


Figure 11. Medicinal Plant Detection result

The web-based identification application developed in this research offers a promising solution to help the public identify medicinal plants indigenous to North Sulawesi Province and understand their benefits in disease prevention and treatment. This program offers a convenient and accessible avenue for individuals to explore the medicinal properties of local plants, which is especially valuable amid rising interest in alternative medicine. This approach can serve as a foundation for future research and development in pharmacology centered on traditional herbal medicine, paving the way for exploring natural remedies as potential avenues for drug development. This is crucial given the increasing demand for safe and efficacious healthcare options.

4. Conclusion

The medicinal plant identification model proposed in this study using CNN has shown excellent and stable performance. Five identification models have been created by performing cross-validation 5 times. Evaluation of the results showed that even though the weakest model, it still provided high validation accuracy, reaching 71.144%, while the best model achieved perfect validation accuracy of 79.104%. The system also managed to achieve high accuracy when testing new data. The weakest model provided a testing accuracy of 73.50%, which is the same as the validation accuracy, while the best model achieved an excellent testing accuracy of 73.00%. In addition, a web-based system has been successfully implemented to use this best model and operates smoothly. Future work can improve the accuracy by hyperparameter tuning and integrating with experts in the web system.

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